Customer Churn Analysis and Prediction

A Data-Driven Approach Using Instacart Customer Behavior

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1. Introduction

Customer retention is a critical driver of long-term profitability for digital platforms. In highly competitive markets such as online grocery delivery, understanding customer purchasing behavior and identifying early signals of churn can provide a significant strategic advantage.

This project analyzes customer transaction data from Instacart to:

- Examine customer purchasing behavior
- Identify key drivers of churn
- Segment customers based on behavioral patterns
- Develop predictive models to estimate churn risk

By combining exploratory data analysis, feature engineering, and machine learning techniques, this study provides both analytical insights and actionable business recommendations.

2. Dataset Description

The analysis utilizes the Instacart Market Basket dataset, which contains anonymized transactional data from a large online grocery platform.

Dataset Summary

Component	Description
Orders	~3.4 million orders
Prior Orders	~32 million product purchases
Customers	~206,000 users
Products	~49,000 items
Aisles	134 categories
Departments	21 categories

The dataset enables reconstruction of complete customer purchase histories, making it suitable for behavioral analysis and churn modeling.

3. Methodology

The project follows a structured analytical pipeline:

1. Data preparation and integration
2. Feature engineering
3. Churn definition
4. Exploratory data analysis
5. Customer segmentation
6. Predictive modeling
7. Model evaluation and interpretation

4. Feature Engineering

Customer-level features were constructed to capture purchasing behavior.

4.1 Order Behavior

- Total number of orders
- Average days between orders
- Variability in purchase intervals

4.2 Basket Characteristics

- Average basket size
- Maximum and minimum basket size

4.3 Product Diversity

- Total number of products purchased
- Number of unique products
- Number of unique aisles and departments

4.4 Loyalty Indicators

- Reorder rate
- Total reordered items

4.5 Shopping Patterns

- Average cart position of items

These features provide a comprehensive representation of customer engagement and purchasing habits.

5. Churn Definition

As the dataset does not explicitly define churn, a proxy measure was constructed based on purchasing behavior.

A customer is classified as churned if:

- The average time between purchases exceeds a threshold
- The customer has relatively low order frequency

This approach captures customers who have significantly reduced or ceased activity.

Churn Rate

Approximately 23% of customers were classified as churned.

6. Exploratory Data Analysis

6.1 Customer Distribution

The dataset exhibits a moderate class imbalance, with the majority of customers remaining active.

6.2 Order Frequency

Customer activity is highly skewed:

- Most customers place relatively few orders
- A smaller group of highly active users contributes a significant portion of transactions

6.3 Purchase Timing

Most customers reorder within 7 to 20 days, suggesting recurring weekly or biweekly shopping behavior.

6.4 Basket Size

Typical basket sizes range from 5 to 15 items, with occasional high-volume purchases.

6.5 Reorder Behavior

Active customers tend to exhibit significantly higher reorder rates, indicating product familiarity and habit formation.

7. Customer Segmentation

Customers were segmented using K-Means clustering based on behavioral features.

Segment Summary

Segment	Characteristics	Churn Risk
Segment 0	Low activity, long order gaps	High
Segment 1	Moderate activity, consistent ordering	Very Low
Segment 2	High basket size, diverse purchases	Medium
Segment 3	Highly active, frequent purchases	Very Low

Key Observations

- Segment 0 shows the highest churn risk (~56%)
- Segment 3 exhibits near-zero churn and represents highly loyal customers
- Segmentation effectively distinguishes customer behavior patterns

8. Predictive Modeling

Two classification models were developed to predict customer churn.

8.1 Logistic Regression

A baseline linear model providing interpretability.

Performance:

- Accuracy: ~98%
- ROC-AUC: 0.997

8.2 Random Forest Classifier

A non-linear ensemble model capturing complex relationships.

Performance:

- Accuracy: ~100%
- ROC-AUC: 1.00

9. Feature Importance Analysis

Feature importance from the Random Forest model identified the key predictors of churn.

Top Drivers

1. Average days between orders
2. Total number of orders
3. Variability in order frequency
4. Total reordered items
5. Product diversity

The dominant factor is purchase frequency, indicating that changes in ordering patterns are strong signals of disengagement.

10. Key Insights

10.1 Purchase Frequency is Critical

Customers who delay purchases beyond their typical cycle are significantly more likely to churn.

10.2 Low-Frequency Customers Are High Risk

Customers with few total orders and irregular activity are the most likely to disengage.

10.3 Loyal Customers Show Consistent Behavior

Highly active customers:

- Order frequently
- Maintain high reorder rates
- Exhibit stable purchasing patterns

10.4 Product Loyalty Reduces Churn

Customers who repeatedly purchase the same products demonstrate stronger retention.

11. Business Implications

The findings suggest several actionable strategies:

11.1 Early Churn Detection

Monitor deviations in purchase frequency to identify at-risk customers.

11.2 Targeted Retention Campaigns

- Offer promotions to low-frequency customers
- Incentivize repeat purchases

11.3 Personalized Recommendations

Encourage product loyalty through recommendations based on past purchases.

11.4 Customer Segmentation Strategies

Develop differentiated marketing strategies for each behavioral segment.

12. Limitations

- Churn is inferred rather than explicitly observed
- Potential data leakage due to feature construction

- Lack of demographic or pricing information

13. Future Work

Potential improvements include:

- Time-series modeling of purchasing behavior
- Survival analysis for churn prediction
- Customer lifetime value estimation
- Integration with recommendation systems

14. Conclusion

This project demonstrates how customer transaction data can be leveraged to understand behavioral patterns and predict churn risk.

The results highlight that:

- Purchase frequency is the most important indicator of retention
- Behavioral segmentation provides meaningful customer insights
- Machine learning models can effectively identify at-risk customers

These insights can support data-driven decision-making and improve customer retention strategies.